



Optimization and Lab Characterization of the TIME Instrument

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TIME Background

Tomographic Ionized-Carbon Intensity Mapping Experiment

- Singly ionized carbon emission lines
- Cosmic phase transition ~1 billion years after Big Bang



Recombination → Dark Ages → Reionization

~ 370000 yrs after Big Bang

Cooled through expansion

Condense into neutral hydrogen

Uneven distribution of matter

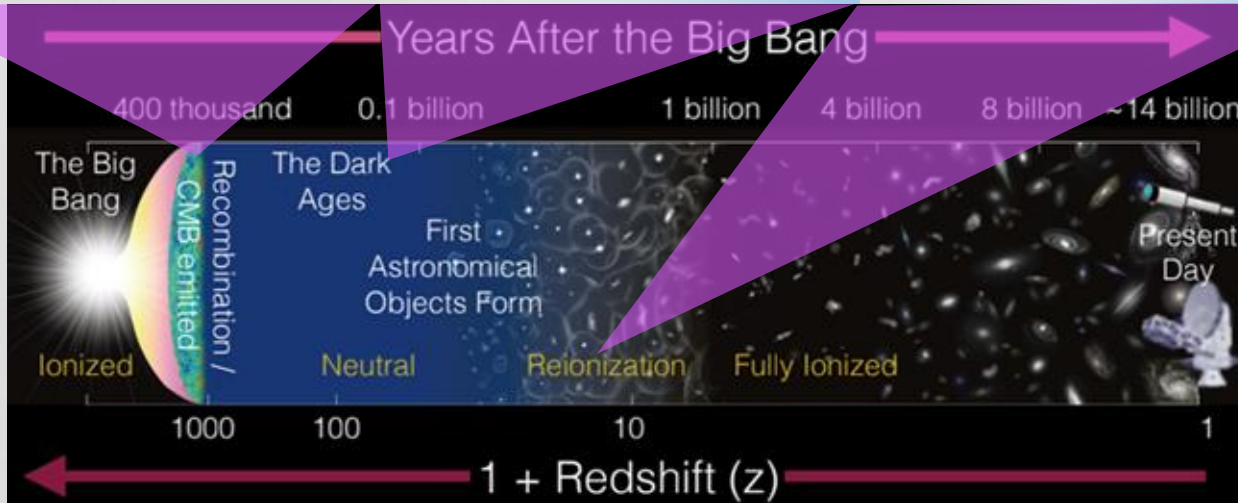
Galaxies and stars begin to form

Ionizing radiation from new galaxies

Neutral hydrogen now ionized

What other sources of ionization?

ALMA Observatory:
<https://www.almaobservatory.org/en/160616-header-web-eng/>



What We Are Building

Cryostat housing detectors and spectrometers.

Will be attached to the bottom of a telescope at Kitt Peak for observation!



Figure 2.43 from Jon Hunacek Thesis, Section 2.5

Figure 6.3 from Jon Hunacek Thesis, Chapter 6



What TIME Does

Examining the [CII] emission line allows us to better understand the phase shifts in the universe.

We don't understand reionization very well!
We could find other sources.

Using [CII] instead of ionized hydrogen allows us to observe a slightly different wavelength than other similar experiments.

Artist rendering of a quasar from NASA JPL
<https://images.nasa.gov/details/PIA16114>



Hubble Image of a star cluster from
NASA GSFC
https://images.nasa.gov/details/hubble-catches-stellar-exodus-in-action_17644845341_o

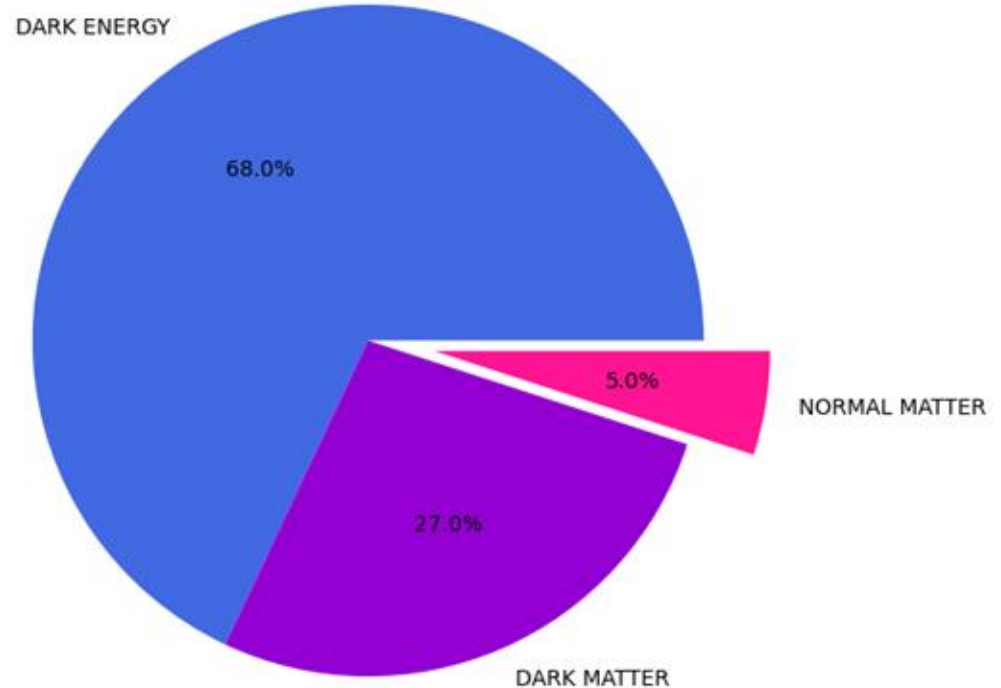


Relevance

Tracing matter as it forms all over the universe.

Could measure wavelengths across all space and time for different forms of matter.

This kind of cosmology allows us to learn more about our history on a cosmic scale!



TES Bolometers

Detectors sit under the spectrometer to receive light split in spatial and spectral axes.

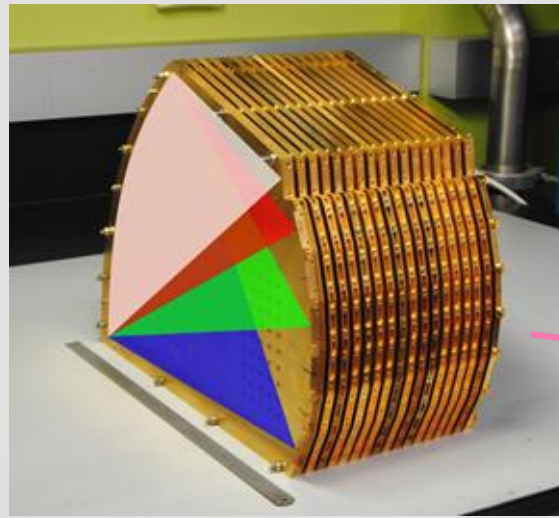


Figure 2.7 form Jon Hunacek Thesis, Section 2.2

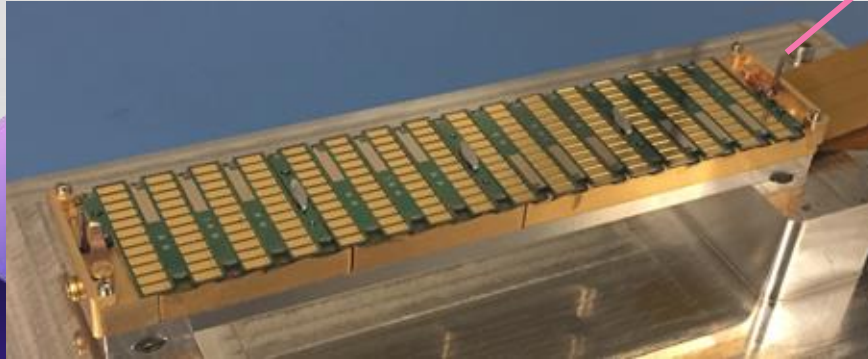


Figure 2.11 form Jon Hunacek Thesis, Section 2.2

spectrometer

detector array

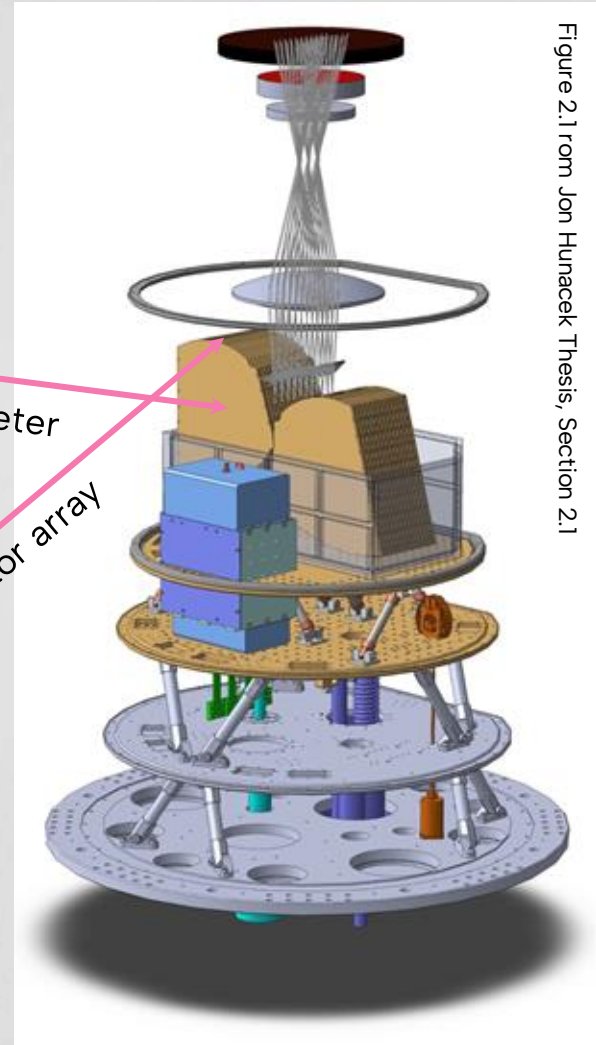


Figure 2.1 from Jon Hunacek Thesis, Section 2.1

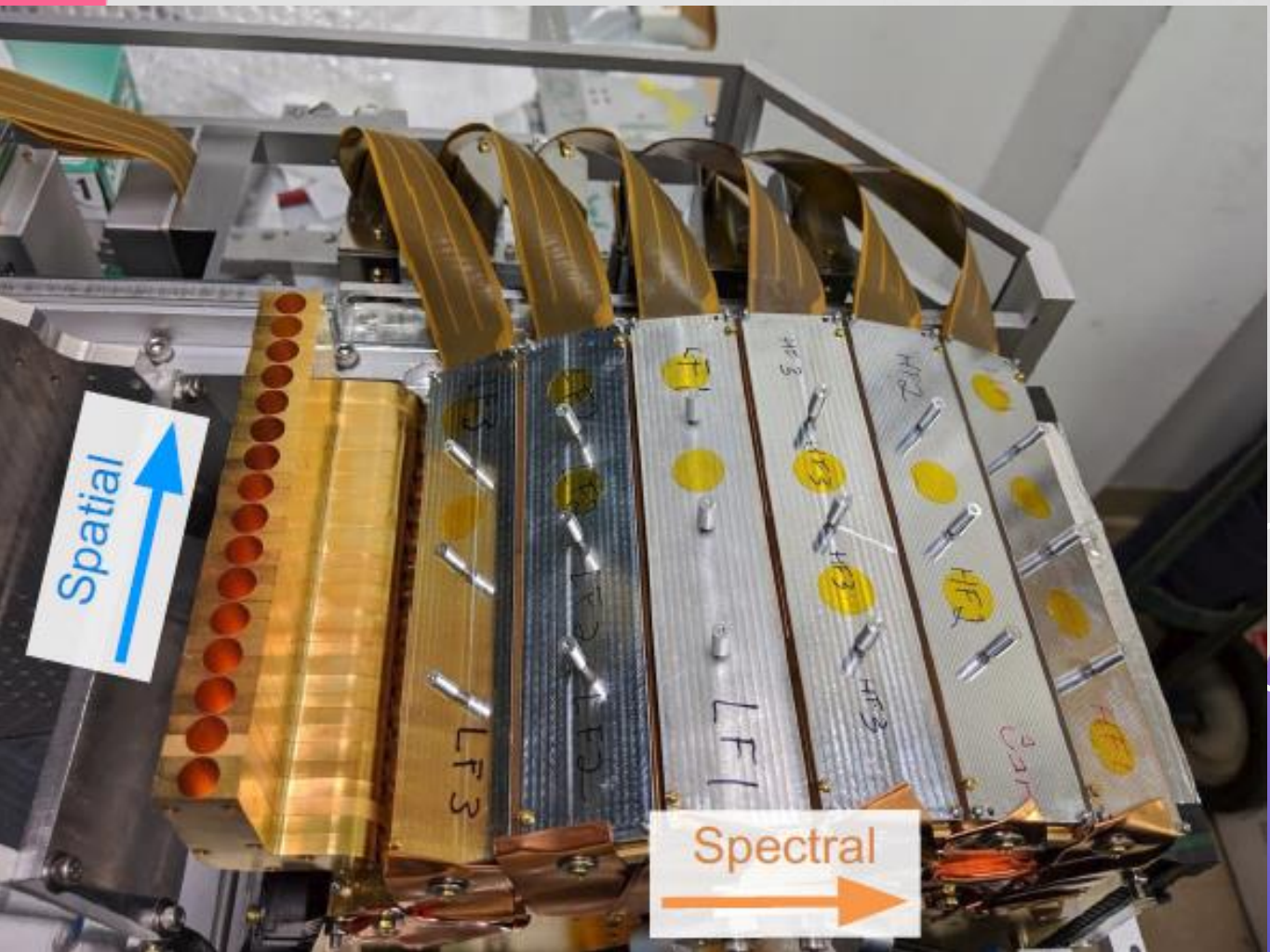
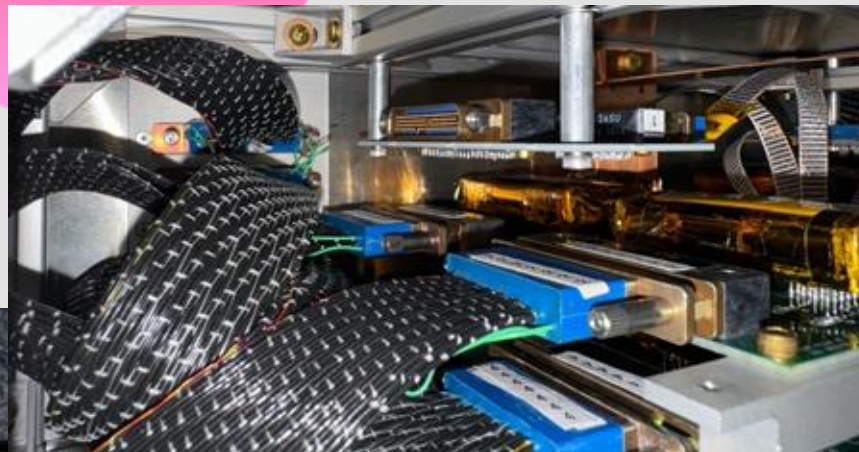


Figure 2.12 from Jon Hunacek Thesis, Section 2.2

The Cryostat

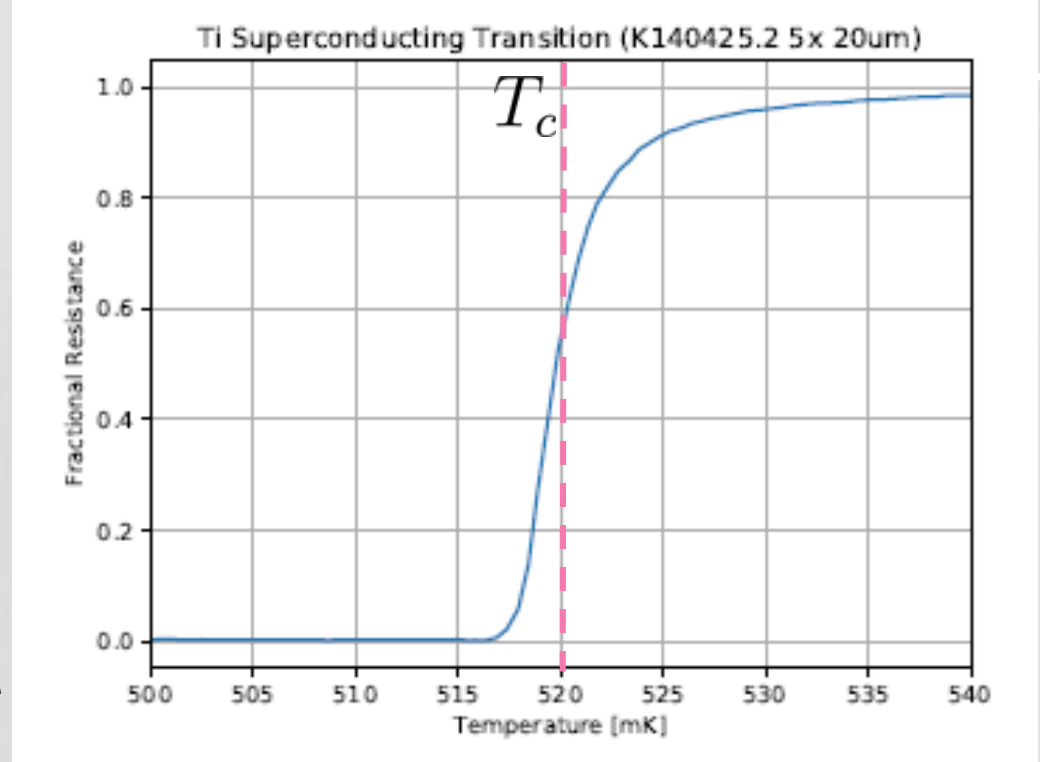


TES Bolometer Design

Film of conductive material that has transition temperature T_c

Small change in temperature causes big change in resistance.

Figure 3.2 from Jon Hunacek Thesis, Section 3.1

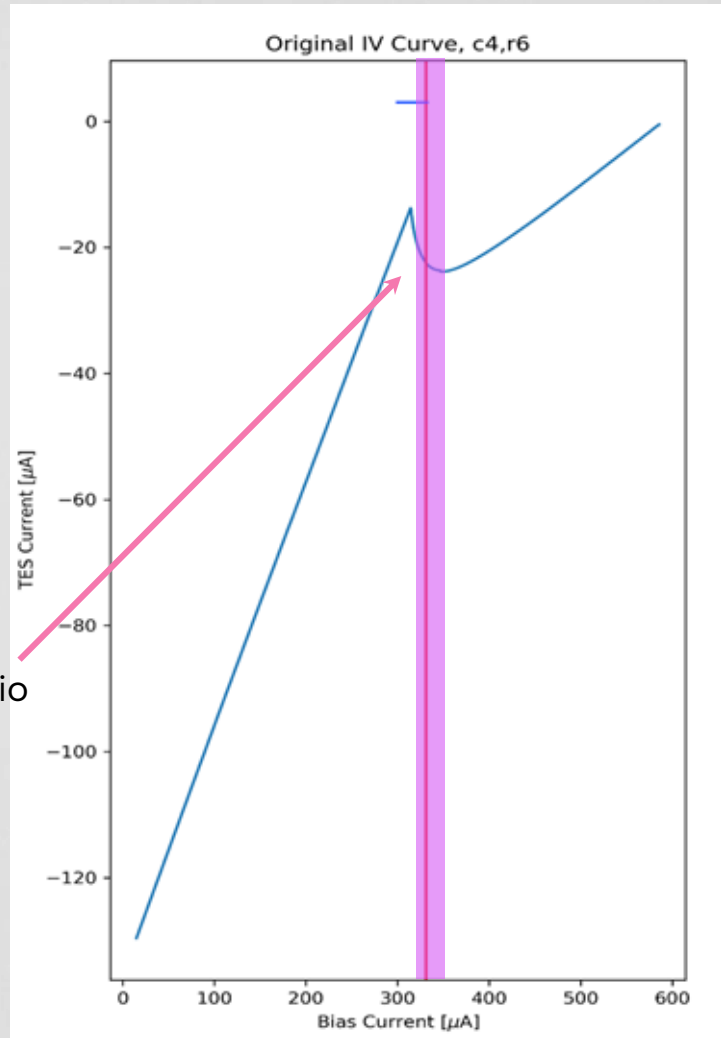


Load Curve Transitions

Load curves compare the current across the detector (TES current) to the input current (Bias current).

The transition on a load curve correlates with the decreasing section of the graph.

Transition

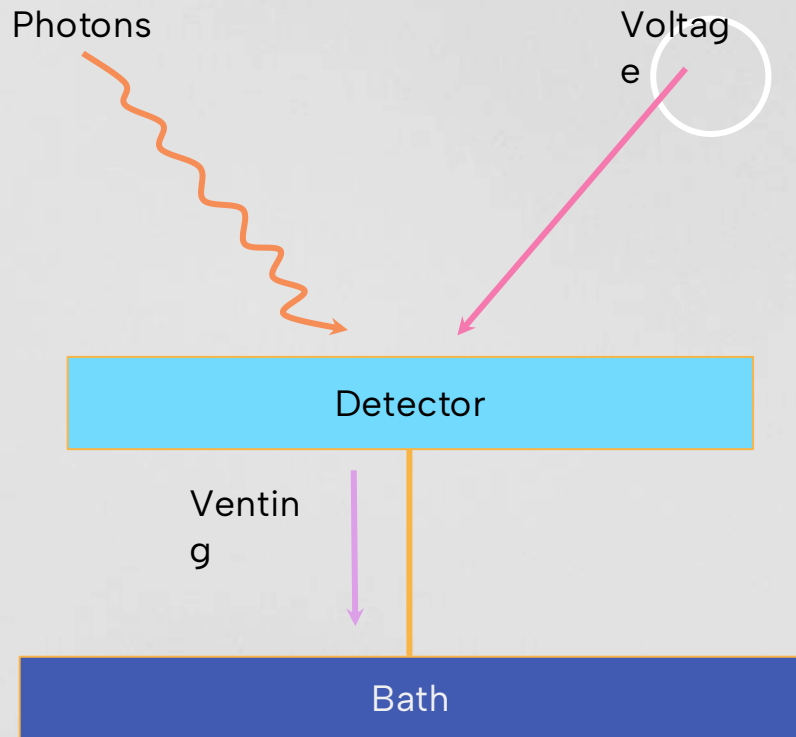


TES Bolometer Methods

Detectors sit on top of a bath that allows for venting of power according to the thermal conductivity

$$P_{bias} + P_{optical} = C \frac{dT}{dt} + P_{bath}$$

$$V_{bias} = I(R_{th} + R_{tes}) + L \frac{dI}{dt}$$



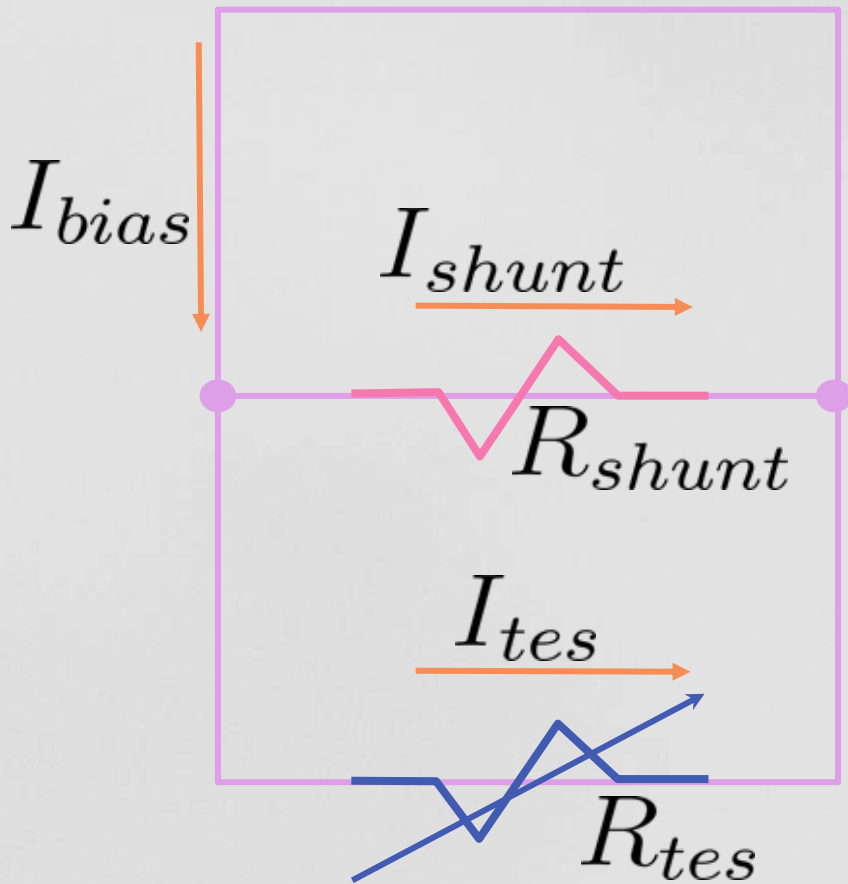
TES Bolometer Design

Detectors connected in parallel
with shunt resistor

Gives constant V_{bias}

$$V_{bias} \equiv I_{bias} R_{shunt}$$

$$I_{bias} = I_{shunt} + I_{tes}$$



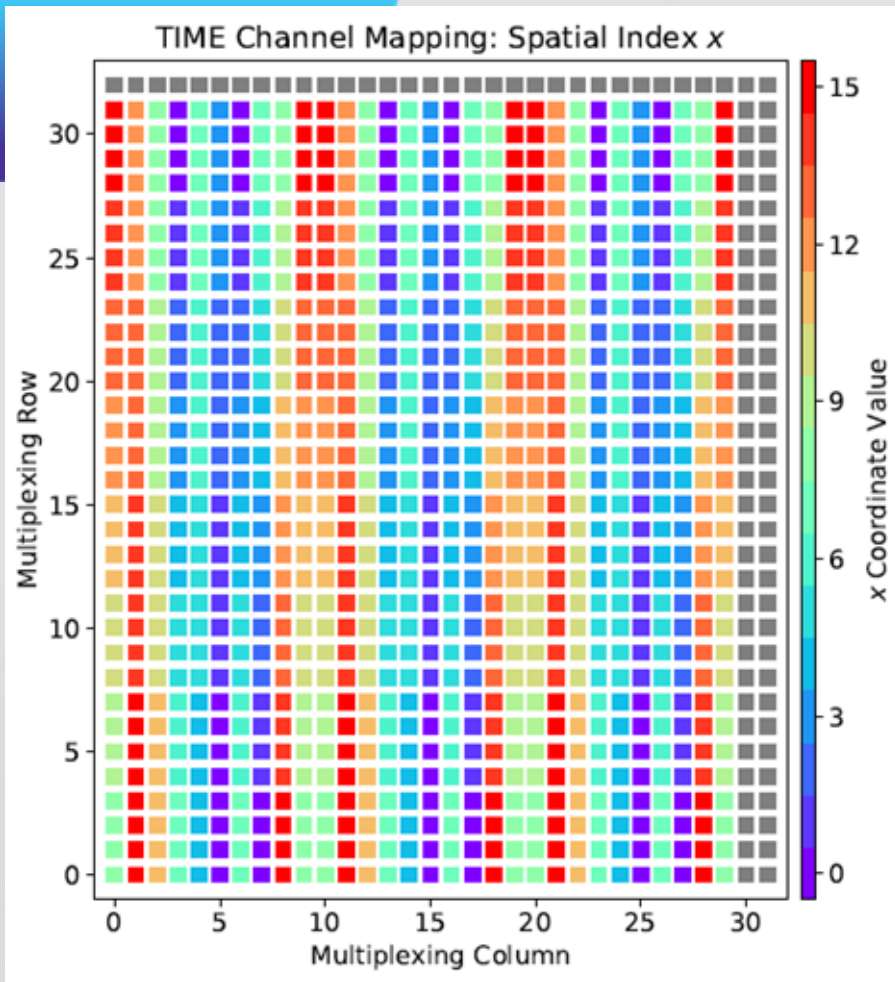
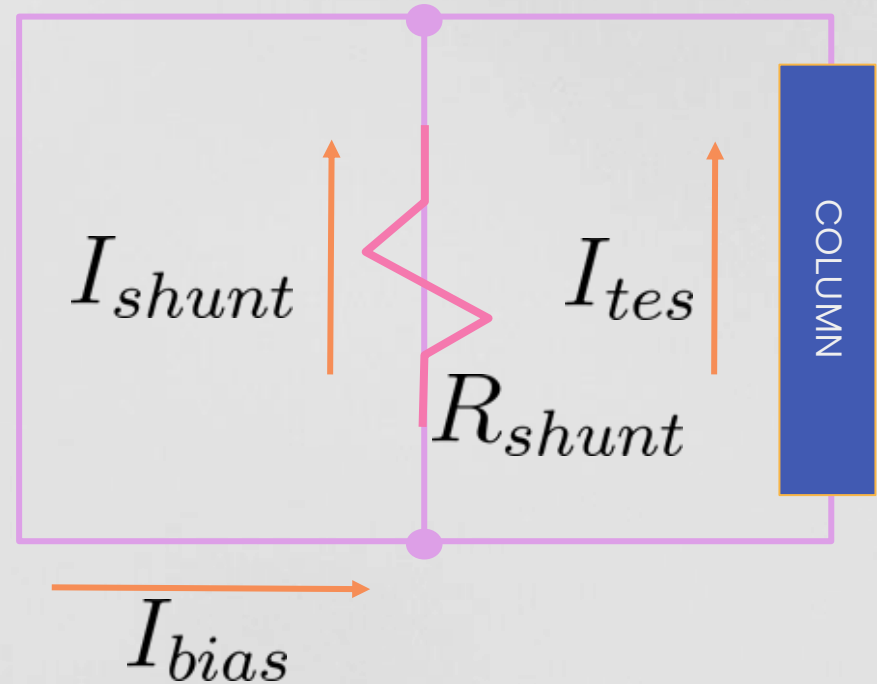


Figure 2.23 from Jon Hunacek Thesis, Section 2.3

Limitations



We can only bias on each column, NOT on an individual detector.



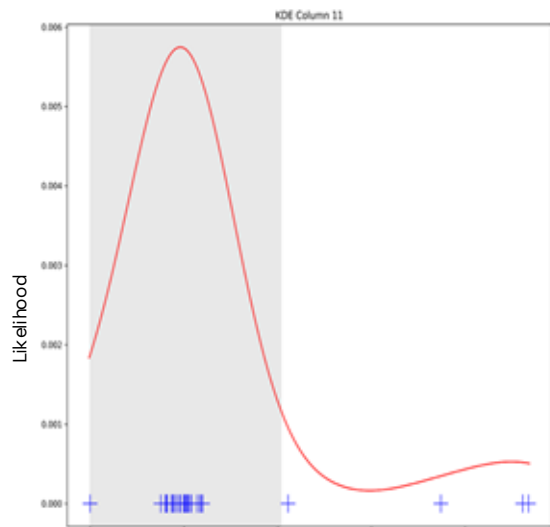
optimal_bias.py

AKA the majority of my time spent!!!

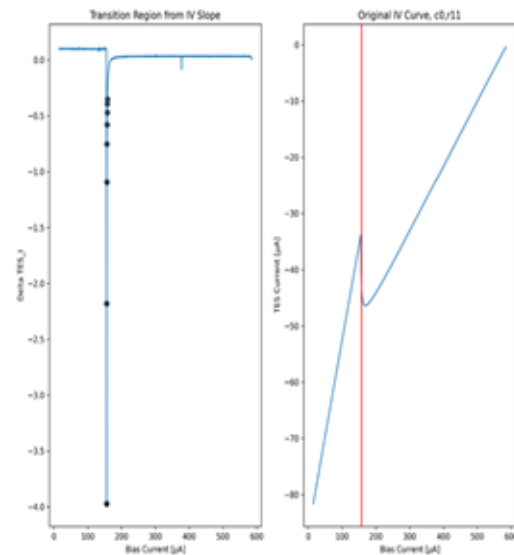
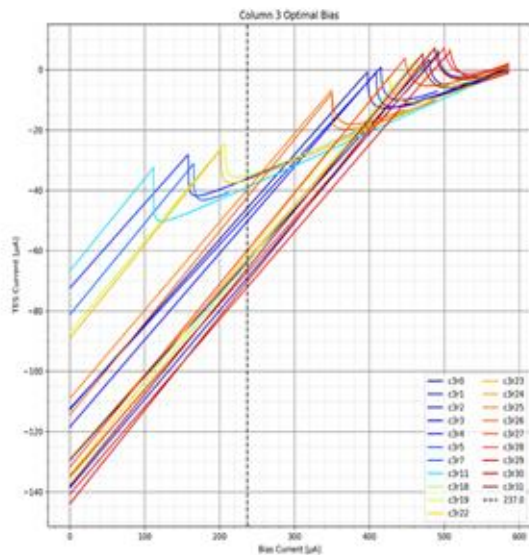
What does it do?

Use the optimal bias script to find the best bias voltage value for each column.

All



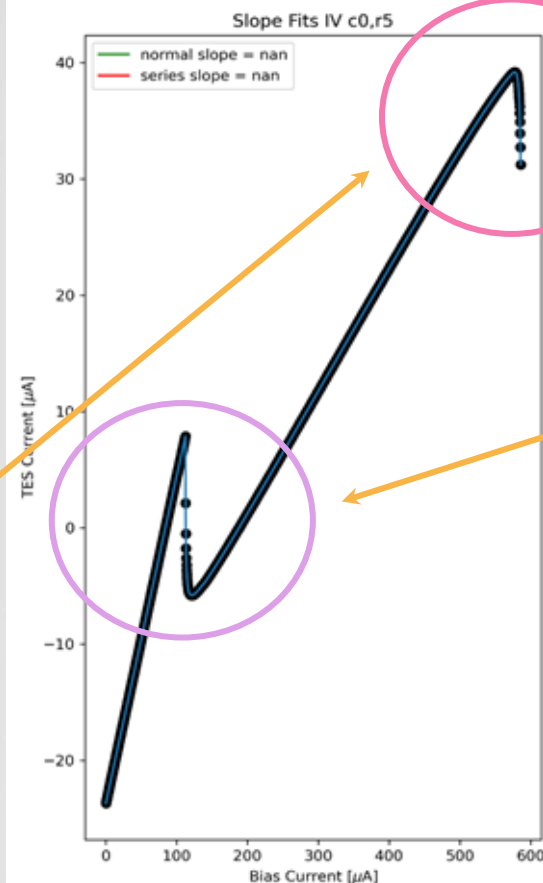
Bias Current



cutoff_finder()

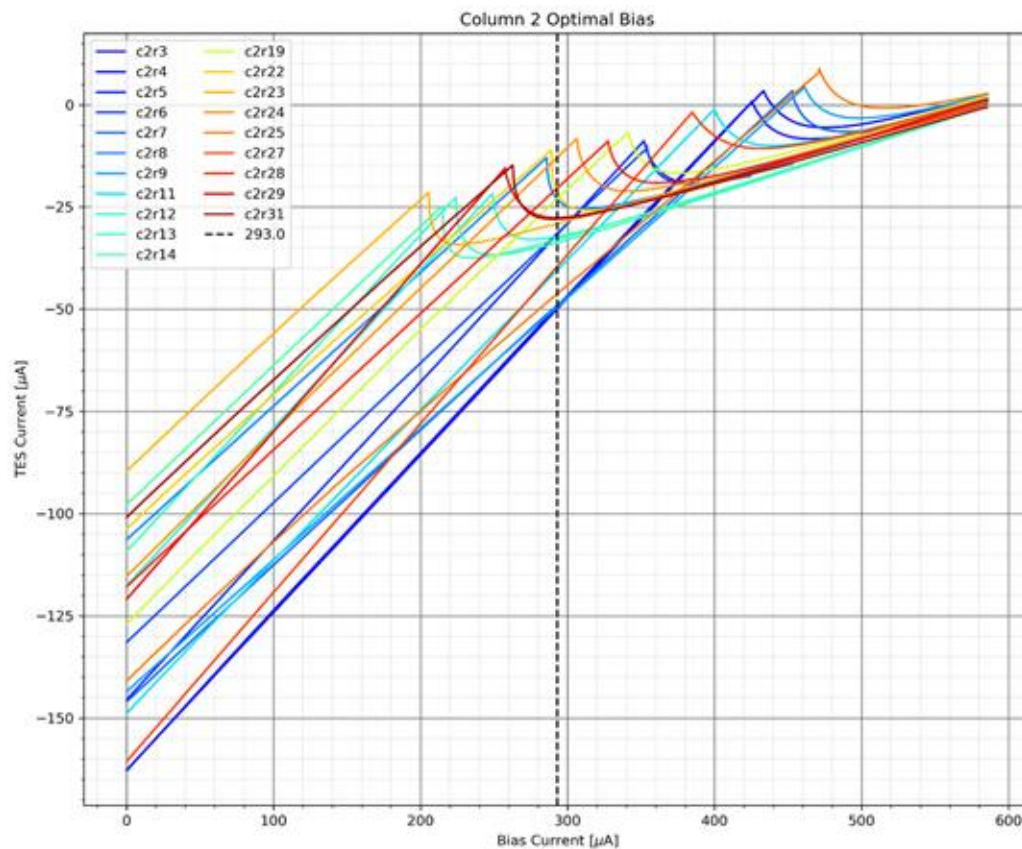
Mitigate multiple transitions due
aluminum or manufacturing defects.

2nd Transition
(BAD!!)



1st Transition
(GOOD!!)

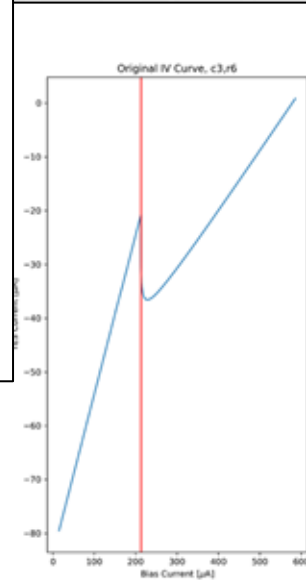
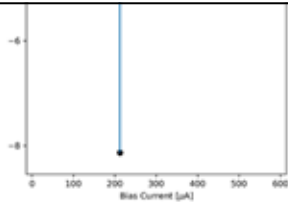
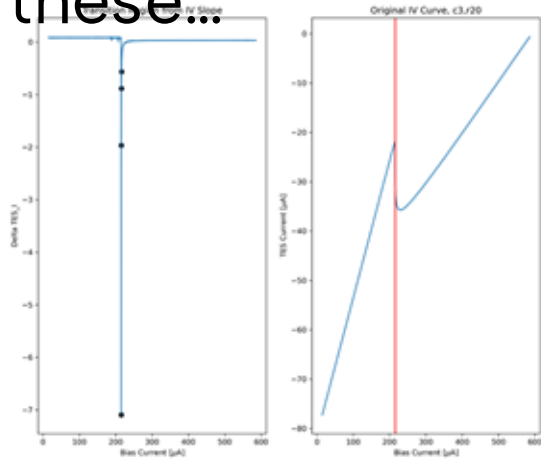
No more funky slope decrease!



col_transition()

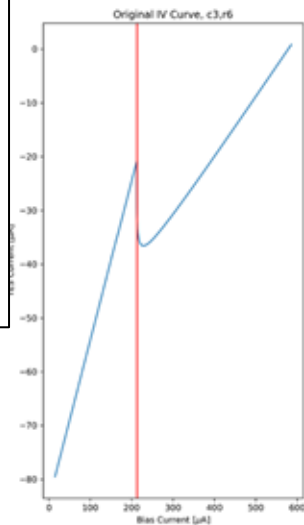
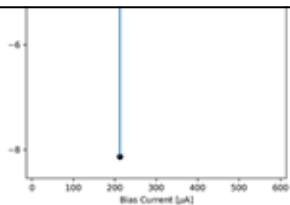
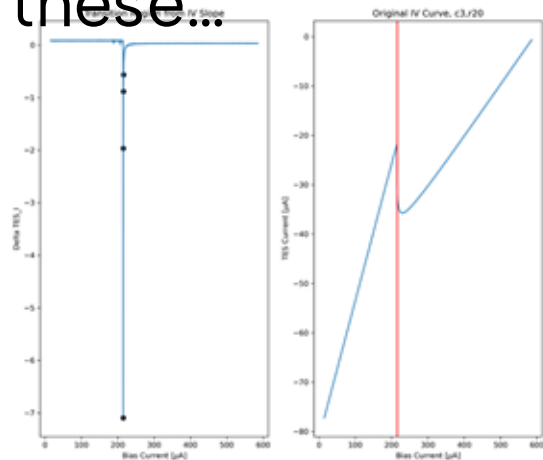
Turn

these...

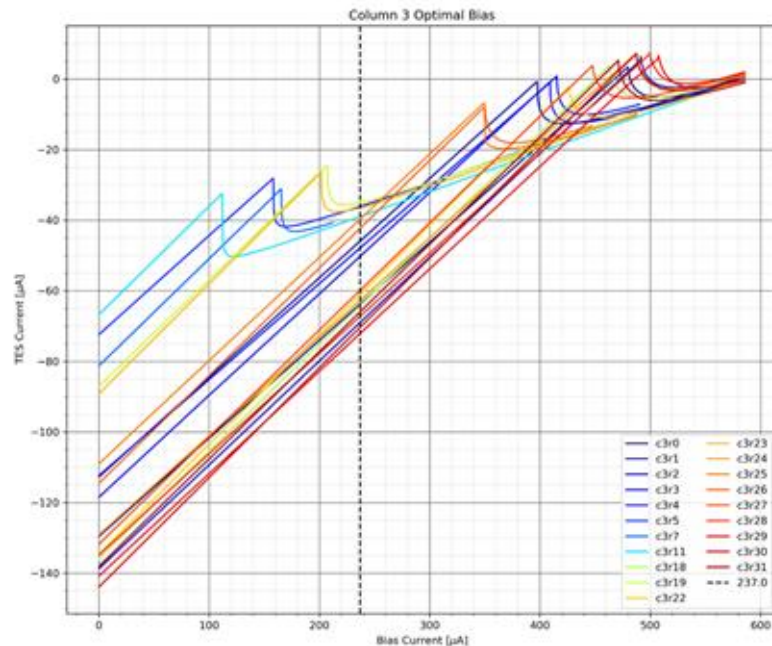


col_transition()

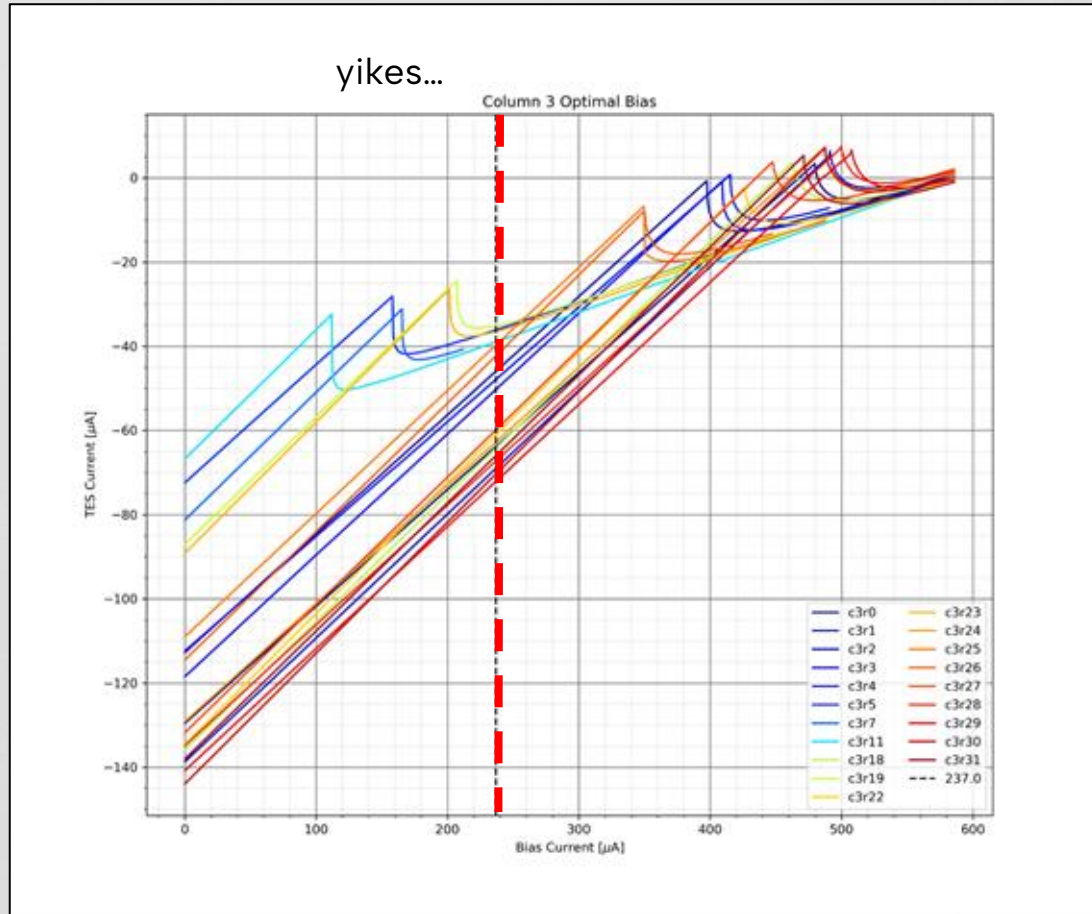
Turn
these...



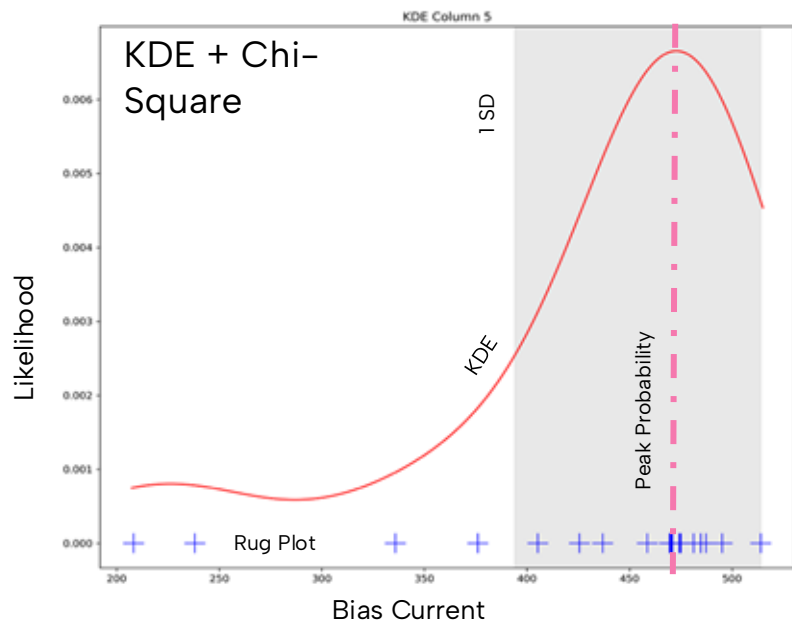
Into this!!



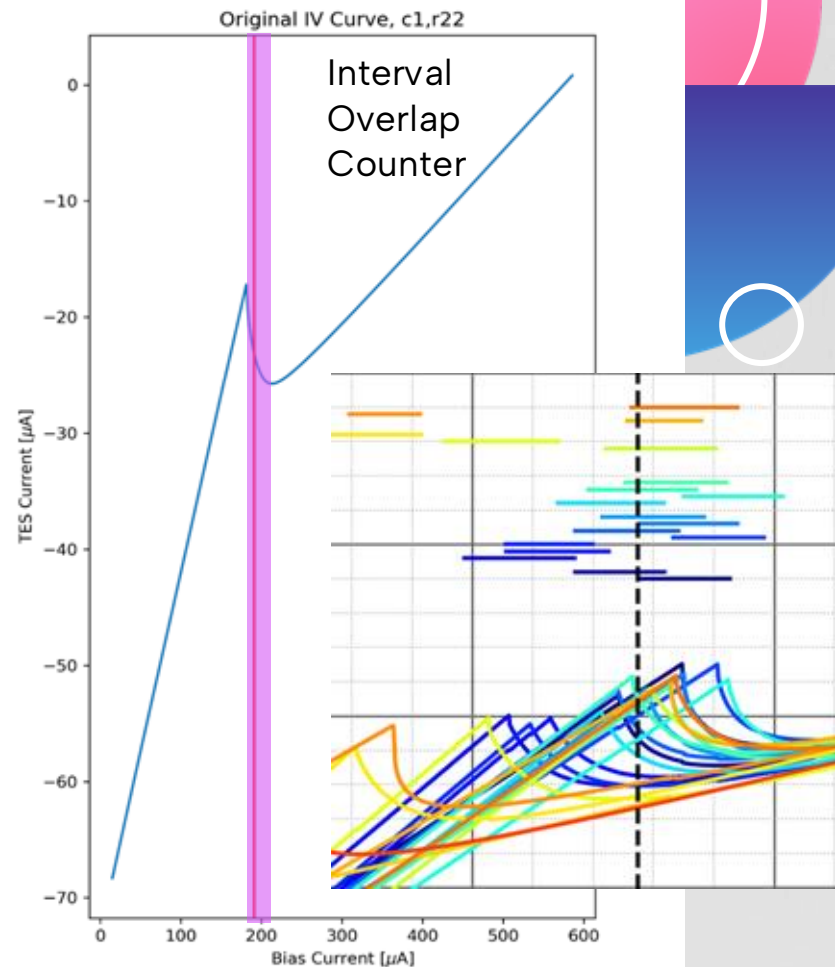
But the transition didn't look right?



How to fix this?



OR



$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

To summarize:



cutoff_finder()

Remove secondary transitions

1

Find individual transitions

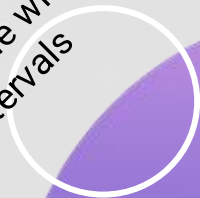
Each detector has an individual bias value!

2

col_transition()

Find the bias value with the most overlapping intervals

3



Next Project: Beam Mapper!

Use IR sources to produce fake data for testing between Kitt Peak visits!



Motor
Controller

Function
Generator



Beam Mapper
Mount

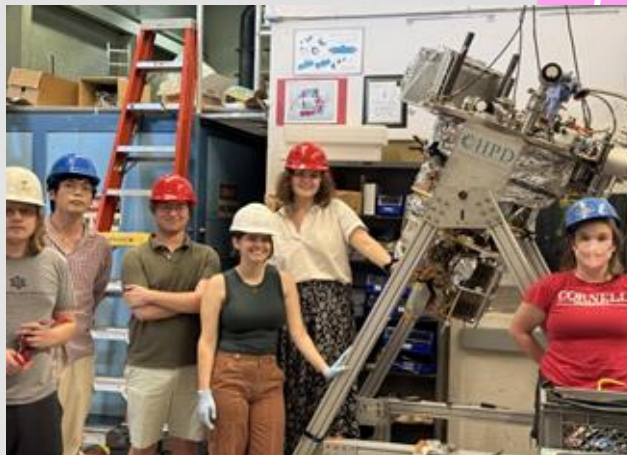
Other Highlights



Polish Party!



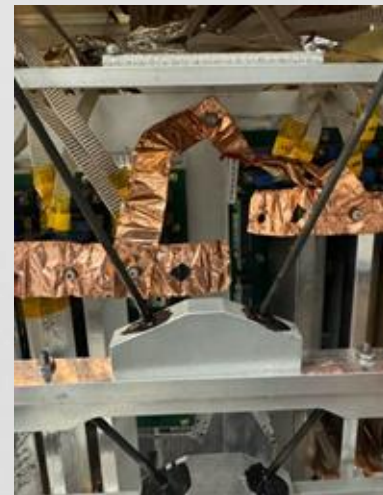
Cryo Disassembly with
10-ton Crane



Tip Test!



Detector Installation
(DO NOT TOUCH!)



Heat syncing



The Future



The Future for TIME

After it all works and lab tests are done, then we take it to Kitt Peak for ~3 years, 3 months per year for observation!!

Cosmology requires long-scale observation.

Future experiments could look at different wavelengths at different times to better understand those phase shifts.



Figure 2.42 from Jon Hunacek Thesis, Section 2.5

The Future for Me

What I learned:

1. I really enjoy the hands-on parts of experimental physics!
1. I like being able to transition between different sub-projects depending on how I feel each day.
1. I find writing code for the experiment fulfilling, but I'm not sure if I would want that as my main thing.
1. I need a lab that is supportive of teaching new people about the project.

Future goals:

1. Explore the more mechanical side of experimental physics.
1. Learn more about subfields adjacent to observational cosmology, maybe in industry.

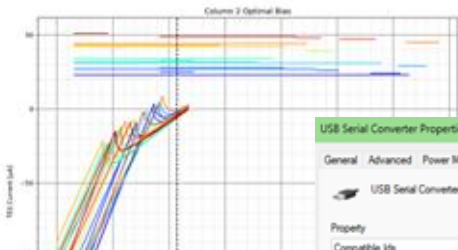


Bonus: Fails

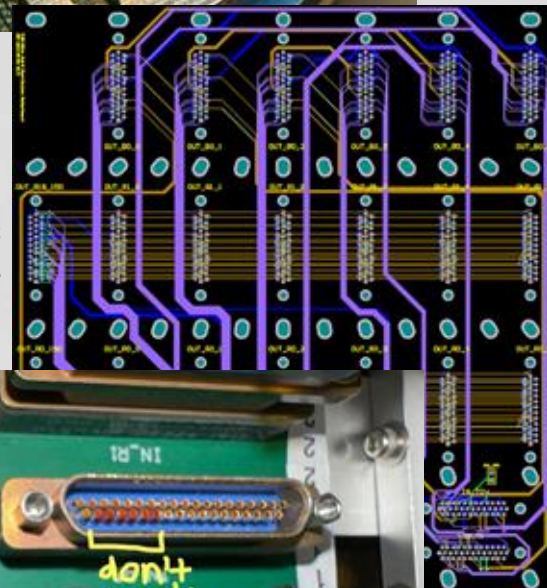
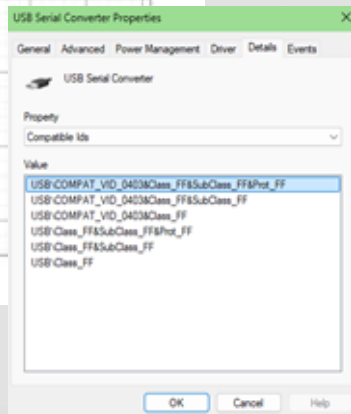
```
### do some iteration to create a list of indices where
max_index = [] #list of indices of local maxes
min_index = [] #list of indices of local mins
other_inflections = [] #idk if we need other inflectio
for i in range(len(diff_tes_i)):
    if diff_tes_i[i] == 0:
        if diff_tes_i[i-1]>0 and diff_tes_i[i+1]<0:
            max_index = max_index + [i]
        elif diff_tes_i[i-1]<0 and diff_tes_i[i+1]>0:
            min_index = min_index + [i]
        else:
            other_inflections = other_inflections + [i]
if len(max_index)>1 and len(min_index)>1:
    return min(max_index[1], min_index[1]) - 1
elif len(max_index)>1:
    return max_index[1]
elif len(min_index)>1:
    return min_index[1]
else:
    return -1
```

```
55 chopper_freq = frequencies[0] * Hz
56 ### setting the first wanted frequency first, then will iterate
57 ### through the rest when doing beammap_take
58 amplitude = 5.0 * V_peak_to_peak
59
60
61 rw = pyvisa.ResourceManager()
62 instr = rw.open_resource('USB0::0x0957::0x1587::MY9001654::0::INSTR')
63
64
65 idn = instr.query('*IDN?')
66 print("found device:")
67 print(idn)
68 # assert '5062122X' in idn, "unexpected device!"
69
70
71 def cmd(s):
72     time.sleep(0.5) # data limit
73     print(s)
74     instr.write(s)
75
76 cmd('*HST')
77
78 print("Turning on...")
```

So. much. code.



```
for i in range(len(tes_i)):
    if i > 100 and i < len(tes_i) - 100:
        slope_before = (tes_i[i]-tes_i[i-100])/100
        slope_after = (tes_i[i+100]-tes_i[i])/100
        if slope_before>0 and slope_after>0:
            max_index = max_index + [i]
        elif slope_before<0 and slope_after<0:
            min_index = min_index + [i]
    if len(max_index)>1:
        return max_index[1] - 1
    #print(max_index[1] - 1)
else:
    return -1
```



When the distribution
board stopped
working for a month
and then fixed itself
T-T



Questions?

Thank you to Victoria Butler for providing the base code and for brainstorming solutions with me! (and for being responsive to my frantic Slack DMs)

Thank you to Abby Crites for being a wonderful PI and being so active in my learning process throughout my time in her lab.

Lastly, thank you to all the other amazing people I worked with this spring and summer!

CREDITS: This presentation template was created by **Slidesgo**, and includes icons by **Flaticon**, infographics & images by **Freepik** and content by **Swetha Tandri**

Sources:

All images without captions are my own.

Appel, J. W., Bennett, C. L., Brewer, M. K., Bustos, R., Chan, M., Chuss, D. T., Cleary, J., Couto, J. D., Dahal, S., Datta, R., Denis, K., Eimer, J., Essinger-Hileman, T., Harrington, K., Iuliano, J., Li, Y., Marriage, T. A., Núñez, C., Osumi, K., ... Xu 徐, Z. (2022). Calibration of transition-edge sensor (TES) bolometer arrays with application to class. The Astrophysical Journal Supplement Series, 262(2), 52. <https://doi.org/10.3847/1538-4365/ac8cf2>

Hunacek, J. R. (2020). TIME: A Millimeter-Wavelength Grating Spectrometer Array for [CII]/CO Intensity Mapping (thesis). <https://thesis.library.caltech.edu/13683/1/thesis-jhunacek-20200427b.pdf>

Kernel density estimation. Kernel density estimation – SciPy v1.15.0.dev Manual. (n.d.). https://scipy.github.io/devdocs/tutorial/stats/kernel_density_estimation.html

